PROJECT 04: Employee Payroll and Deduction Analysis

1. Problem Statement

Calculate gross salary from basic pay, allowances, overtime and incentives; apply progressive illustrative deductions and produce a salary statement.

2. Objectives

• Compute overtime pay and gross salary.
• Apply progressive deduction bands.
• Calculate net salary.
• Print an itemized salary statement for each employee.

3. Assumptions

• Overtime rate is 1.5 times basic hourly rate; monthly hours for hourly conversion are assumed to be 160.
• Deduction is 5% up to Rs.25,000; Rs.1,250 plus 10% of the portion above Rs.25,000 up to Rs.50,000; above that, Rs.3,750 plus 15% of the portion over Rs.50,000.
• These are example deduction rules, not official tax advice.
• Basic pay, allowances, incentive and overtime hours must be non-negative.

4. Algorithm

1. Read employee salary components and overtime hours.
2. Validate the inputs.
3. Calculate overtime pay from basic pay and overtime hours.
4. Add basic pay, allowance, overtime and incentive to get gross salary.
5. Calculate progressive deduction based on gross salary.
6. Subtract deduction from gross salary and display the statement.

5. Flowchart

6. Pseudocode

```
START
  FOR each employee
    VALIDATE all salary inputs
    overtime = hours * (basic / 160) * 1.5
    gross = basic + allowance + overtime + incentive
    IF gross <= 25000: deduction = 5% of gross
    ELSE IF gross <= 50000:
      deduction = 1250 + 10% of (gross - 25000)
    ELSE:
      deduction = 3750 + 15% of (gross - 50000)
    net = gross - deduction
    DISPLAY salary statement
END
```

7. Python Code

```
employees = [
    {"name": "Kiran", "basic": 25000, "allowance": 5000, "overtime_hours": 8, "incentive":
2000},
    {"name": "Sita", "basic": 40000, "allowance": 8000, "overtime_hours": 5, "incentive": 3500},
]
```

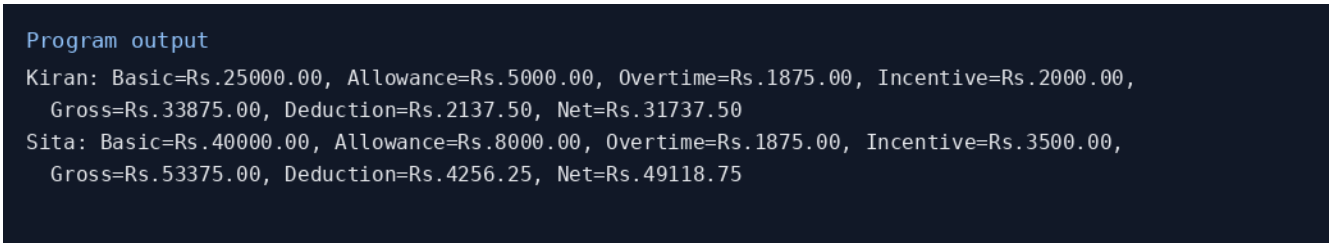
```
def deduction(gross):
    if gross <= 25000:
        return gross * 0.05
    if gross <= 50000:
        return 1250 + (gross - 25000) * 0.10
    return 3750 + (gross - 50000) * 0.15

for e in employees:
    if min(e["basic"], e["allowance"], e["overtime_hours"], e["incentive"]) < 0:
        print(f"{e['name']}: Invalid salary input")
        continue
    overtime = e["overtime_hours"] * (e["basic"] / 160) * 1.5
    gross = e["basic"] + e["allowance"] + overtime + e["incentive"]
    deduct = deduction(gross)
    print(f"{e['name']}: Basic=Rs.{e['basic']:.2f}, Allowance=Rs.{e['allowance']:.2f},
Overtime=Rs.{overtime:.2f}, Incentive=Rs.{e['incentive']:.2f}, Gross=Rs.{gross:.2f},
Deduction=Rs.{deduct:.2f}, Net=Rs.{gross-deduct:.2f}")
```

8. Test Cases

Test	Input / condition	Expected result
Low gross salary	Gross Rs.20,000	Deduction is 5% = Rs.1,000
Middle band	Gross Rs.40,000	Deduction = Rs.1,250 + Rs.1,500 = Rs.2,750
Upper band	Gross Rs.60,000	Deduction = Rs.3,750 + Rs.1,500 = Rs.5,250
Negative input	Allowance is -100	Employee record is rejected

9. Output Screenshot

The following screenshot-style image was generated from the captured output of the program execution.



Captured output

```
Kiran: Basic=Rs.25000.00, Allowance=Rs.5000.00, Overtime=Rs.1875.00, Incentive=Rs.2000.00,
Gross=Rs.33875.00, Deduction=Rs.2137.50, Net=Rs.31737.50
Sita: Basic=Rs.40000.00, Allowance=Rs.8000.00, Overtime=Rs.1875.00, Incentive=Rs.3500.00,
Gross=Rs.53375.00, Deduction=Rs.4256.25, Net=Rs.49118.75
```

10. Conclusion

The program computes gross and net salary transparently and demonstrates progressive deductions using clearly documented example bands.

Academic note: The program was executed with the sample data shown in the source code. Policies, rates, thresholds and exchange rates are illustrative and should be adjusted to the assignment's official requirements.